

CSE310: Compiler Sessional, January 2026 Term

ICG Online Assignment Practice

Compound Assignment Operators

Time: 40 Mins

Total Marks: 10

Problem

In this online assignment, you are required to extend the grammar to support compound assignment operators (`+=`, `-=`, `*=`, `/=`). Thus, your program should support the following syntax.

```
int a; a = 10; a += 5;
```

Task

- Extend the current grammar to support the new compound assignment operators. Assume there will be no syntax or semantic errors in the given input program and you need to work only with integer variables.
- Update your ICG code to generate assembly code for this new rule/syntax. The target variable should be both the source of the initial value and the destination of the final result. The assembly code output by your program must correctly be assembled with 'fasm'.

Mark Distribution

- Extending the grammar to support compound assignments. (4)
- Code generation for compound assignments. (6)

Sample Input-Output

Input Program	Generated Sample Assembly (print library omitted)
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<pre> int main() { int a; a = 10; a += 5; println(a); a *= 2; println(a); return 0; } </pre>	<pre> main: PUSH EBP MOV EBP, ESP SUB ESP, 4 MOV EAX, 10 MOV [EBP-4], EAX MOV EAX, [EBP-4] MOV EDX, EAX MOV EAX, 5 ADD EAX, EDX MOV [EBP-4], EAX MOV EAX, [EBP-4] CALL print_number MOV EAX, [EBP-4] MOV ECX, EAX MOV EAX, 2 IMUL ECX MOV [EBP-4], EAX MOV EAX, [EBP-4] CALL print_number MOV EAX, 0 JMP .L1 .L1: ADD ESP, 4 POP EBP RET </pre>
Expected Output	<pre> 15 30 </pre>